

Computing and the Arts: Establishing Theoretical and Methodological Foundations for Cross-Disciplinary Collaboration

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Abstract

The last five years have resulted in substantial changes to how computing affects work, how work affects computing, and how work and computing operate in tandem to affect society. From advances in automation, artificial intelligence, and virtual/extended reality, to the entrenchment of hybrid and remote work arrangements, and the documented harmful societal impacts that computing work has produced, these changes to computing-work relationships raise concern *and* opportunities to reimagine these relationships in new ways. CSCW has an opportunity and a responsibility to ensure that the kinds of futures we imagine and enact benefit workers, communities, and future generations. Artistic research is well-positioned to help us not only understand, but imagine new pathways forward in response to pressing CSCW questions. By hosting a panel of experts in artistic methods well-equipped to help us imagine these futures, we expect to lay the groundwork for mutually respectful cross-disciplinary collaboration between arts and computing that makes more space in our field for different kinds of thinking, approaches to problems, and new imaginaries.

CCS Concepts

• **Computing methodologies**; • **Theory of computation**; • **Social and professional topics** → **Computing profession**;

Keywords

Arts, Methods, Computing, Imaginaries, Cross-Disciplinary

ACM Reference Format:

Angela Schöpke-Gonzalez, Kellie Dunn, Shaowen Bardzell, Makayla Lewis, Maria Murray, and Catherine Wiczorek. 2025. Computing and the Arts: Establishing Theoretical and Methodological Foundations for Cross-Disciplinary Collaboration. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen,

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CSCW Companion '25, Bergen, Norway

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ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3748283>

Norway. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3715070.3748283>

1 Introduction

CSCW studies relationships between computing and work: how computing affects work, how work affects computing, and how work and computing operate in tandem to affect society. The last five years have resulted in notable changes to all of these relationships. For example, the proliferation of new technological possibilities through automation [27], artificial intelligence [20], and virtual/extended reality [17] have affected how we think about workers' roles in relation to computing technologies. Traditional norms about where and how we work have fluxed dramatically since the onset of the global COVID-19 pandemic with an increase in hybrid, distributed, and remote work [3, 9]; ongoing digital nomadism [23]; platform and gig work [36]; and many more changes to conventional work arrangements. Finally, the work that computing and related disciplines produce has come under greater scrutiny for the harms that algorithm design, implementation, and application can perpetuate [4, 5, 25, 26].

These changes raise opportunities for concern *and* for studying and reimagining relationships between computing and work. CSCW has an opportunity and a responsibility to ensure that the kinds of futures we enact benefit workers, communities, and future generations. While seeking to share the stage with large technology companies in future-imagining and future-building is not easy, it is essential that CSCW draw on its expertise to offer alternative, more equitable, just, and sustainable futures. As ethnographers Hayes et al. write, "There will not be one best image of society, because the imagination thrives on multiplicity and diversity and the opening of possibilities" [15]. To meet this challenge, we invite the especially well-positioned expertise of artistic researchers to help us understand and imagine new pathways forward in response to pressing CSCW questions. As science fiction author Ursula K. Le Guin writes:

"I think hard times are coming when we will be wanting the voices of writers [and artists, more generally] who can see alternatives to how we live now, and can see through our fear-stricken society and its obsessive technologies to other ways of being, and even imagine some real grounds for hope. We will need writers

who can remember freedom, poets, visionaries – the realists of a larger reality.” [18]

By hosting a panel of experts in artistic methods well-equipped to help us imagine these futures, we expect to lay the groundwork for mutually respectful cross-disciplinary collaboration between arts and computing that makes more space in our field for different kinds of thinking, approaches to challenges, and new imaginaries.

2 Proposed Structure and Format

This panel will articulate the kinds of value that the arts offer computing and that computing offers the arts, challenges and risks of collaboration between computing and arts disciplines, and provide initial resources on how to enact mutually respectful collaborations. We propose a traditional panel format, including speakers with a diversity of expertise who are prepared to discuss the topic with the facilitator, with each other, and with the audience. We would begin with a brief introduction of the panelists and clarification of what the discussion will be about, and then proceed to a facilitated panel discussion following a prepared list of prompts, concluding with the opportunity to take questions from the audience and panelists. We will share resources and information generated by the panel with all attendees in the form of a website, document, or other accessible digital resource.

2.1 Articulating Value

From their positions of expertise as arts and computing professionals, panelists will discuss the types of knowledge and value that the arts offer. For example: reflexive insights [11, 30, 32]; ability to perceive the world in different and new ways [10, 33]; turning relational [2, 37], care [24], feminist technoscience [1, 8], and new materialist [14] theories into practice; slowing down [16, 22, 31]; and imagining futures and setting directions for computing disciplines [11, 13, 19, 21, 34]. As panelists discuss the values that the arts offer, they will also be invited to critically reflect on what artists may have to gain by collaborating with computing professionals: what value can the CSCW community offer the arts? How can we ensure mutual benefit?

2.2 Identifying Challenges and Risks for Collaboration

Given the arts’ epistemologically and economically marginalized position, extractive computing practices may put artists’ financial well-being at risk. For example, recent trends in tech companies using artists’ copyrighted materials to train large language models (LLMs) without compensation or recognition [12, 28] have affected artists’ livelihoods [7, 35]. Computing practices also change how artistic work happens and indeed whether it happens at all. We will invite panelists to reflect on how CSCW can respond to this challenge by imagining computing futures that work *with* artists.

Accompanying very real financial risks, artists face computing disciplines not taking the knowledge that the arts produce seriously. While computing disciplines have often viewed the arts as producing valuable products – e.g., using the arts in science communication to tell sciences’ stories – they have struggled to take seriously the unique *knowledge* that arts processes can produce [6, 29]. We will invite panelists to discuss: how can CSCW researchers who

are or are collaborating with artists plan their research to learn as much as possible from the practice of art, beyond its products?

Finally, we will invite panelists to reflect on their experiences of doing arts-based research in computing: what are the biggest challenges they have faced in doing arts-based research in computing? What do they wish CSCW researchers understood about artistic research? What are the barriers to doing (and publishing) artistic research in CSCW?

2.3 Generating Resources for Next Steps

Both arts and computing disciplines can struggle to communicate with one another. Each discipline (and individual within that discipline) uses different language, works on different time scales, and has different priorities. Arts and computing disciplines working together means a very real communication challenge. With a view to generating resources that computing professionals can reference when considering work with artists or arts-based methods, we will invite panelists to reflect on discussion prompts such as the necessity of an arts background, engaging artists as collaborators, and respecting differences in practice without tokenizing. We will summarize responses to these questions and make them available via a publicly accessible website after the conclusion of the conference.

3 Technical Requirements

Seating, microphone equipment for all panelists and the facilitator, and a microphone for taking audience questions. Recording equipment to capture information for resources for next steps.

4 Biographical Sketches

Dr. Angela Schöpke-Gonzalez (Organizer) is a choreographer and computing professional who draws on arts-based methods to address some of algorithms’ documented ability to perpetrate social, political, and economic harms by updating workflows and methods used in algorithm design, implementation, and application in order. Her work has been funded by University of Michigan’s ArtsEngine, Center for World Performance Studies, and the School of Information among other institutions.

Kellie Dunn (Organizer) is a PhD fellow at the University of Copenhagen, Department of Computer Science, Human Centred Computing section. Her research, which is part of the ReWork: futures of hybrid work project funded by Digital Innovation Centre Denmark, explores art-research-industry collaborations and the role of artistic practice in research, investigating how these interdisciplinary exchanges can inform theoretical understandings and design of cooperative work technologies and systems.

Dr. Shaowen Bardzell (Panelist) is a Professor in the School of Interactive Computing at Georgia Institute of Technology, where is also the School chair. Her work augments contributions of feminism, design, and social science to support technology’s role in social change. She is interested in feminist art, women’s artmaking, and the bodies.

Dr. Makayla Lewis (Panelist) is a Senior Lecturer in Computer Science (User Experience Design) at Kingston University London, UK. Her research focuses on creative and first person human-computer interaction, accessibility, user experience, design thinking, accessible arts and art spaces, generative AI in the arts,

and human factors in cybersecurity. She has a strong interest in sketching as a design method and has shared her work widely through academic publications, workshops, courses, and blogs. In November 2024, she co-authored the undergraduate and postgraduate textbook *Sketching in Human-Computer Interaction*, published by Springer.

Dr. Maria Murray (Panelist) is an artist/educator/researcher in the Crawford College of Art and Design, Munster Technological University. She is predominantly interested in the application of Socially Engaged Art approaches to participatory design in HCI and CSCW with the intention of disrupting normative assumptions, increasing agency and democracy and imagining alternative computing and social realities, particularly in pressurized and peripheral community settings.

Catherine Wieczorek (Panelist, she/her) is a designer and researcher. She is currently a PhD student in Human-Centered Computing at Georgia Tech. Her dissertation investigates how temporality shapes systemic inequities across human and non-human actors. In parallel, her creative practice engages with climate data as inherently partial and unstable. Through speculative visualizations, generative AI, and print media, she challenges the authority of archives and interrogates the narratives data construct about nature and society.

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